

Enhancing Semantic E-Government Workflows Through Service Oriented Knowledge Provision

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Abstract—This paper follows two goals, first, to provide an insight on the research in the area of semantic workflows for public administrations, and second to present an on-demand service oriented approach on knowledge provisioning for actors in the E-Government processes. Results of the research in the area of semantic workflows will be presented in application scenarios of two EC E-Government research projects, whereas the knowledge provisioning approach will be presented conceptually within an EC E-Learning research project, and as an initial prototype in the EC Business Grids research project.

Keywords— E-Government, Workflows, Knowledge Bus, SOA, Adaptivity, Interoperability

I. INTRODUCTION

After being a buzzword for a long period of time, in the last years E-Government has become something that almost everyone is confronted with. This on one hand has formal reasons that lie in the different cross border initiatives such as the i2010 initiative [1] through the EC, but also because of different national initiatives (e.g. Digitales Österreich in Austria [2], E-Government 2.0 in Germany [3], e-GIF in United Kingdom [3], etc.). Practical aspects fostering the acceptance of E-Government were encouraged first due to trend reversal where the initially main goal in making offline processes available online without adapting them to the changing environments was replaced with the approach where re-engineering of the E-Government processes [5] took place based on the requirements and demands claimed by both sides of the user layer – employees of the government agencies providing- and the end users requesting services. Further development in this direction was the step to provide adaptable and smart E-Government processes based on the dynamically changing environment (e.g. laws, regulations, etc.) and identified requirements based on the everyday work and experiences from the users of the service delivery systems. Finally based on the acknowledgements from the aforementioned challenges, the issue of providing the knowledge required for supporting the end users carrying out knowledge intensive tasks in E-Government, resulted in the research in direction of knowledge provision based on the SOA approach, providing on-demand and correct information to the requesting person.

Following the introductory chapter the structure of the paper is as follows: in the second chapter a brief overview of

the related work on E-Government process management involving semantic technologies is presented. The third chapter presents the used technologies, implementation and deployment scenarios of the semantically enriched E-Government workflows in two EC E-Government research projects. The fourth chapter is dedicated to the application of the SOA approach for knowledge provision by introducing the conceptual research and a prototype deployment scenario in EC research projects. The last chapter provides a summary and brief outlook on further research on the identified challenges.

II. RELATED WORK

As already mentioned in the previous chapter E-Government as an area, and provision of the public services to the citizens as the main aspect, have been focus of many international and national research projects (e.g. OneStopGov [6], BRITE [7], etc.) resulting in advances in different categories. The major issues tackled, being relevant for the work presented in this paper, are situated in the area of providing solutions for highly complex and change-sensitive environments, such as E-Government, taking into account the fact that the solution requires involvement and cooperation of different actors (e.g. citizens and authorities), different application systems (online, offline, distributed, etc) and the aforementioned legal requirements (e.g. laws)[8]. Due to its change-sensitive environment E-Government process management is on one side extremely resource-intensive having to consider all involved parties and circumstances, and on the other side still inflexible or cost-intensive when it comes to react to changes.

This issue was tackled by the introduction of semantic technologies in E-Government in order to provide improvements in the following areas:

- Personalization – one of the most important factors when trying to increase the overall acceptance and usage of the E-Government service delivery channels. Personalization reflects the highest level of sophistication of public online services (i2010, [39]). Successfully realized personalization of the access to the services provides ability to offer end users with made-to-fit E-Government services thus, lowering the usage cost and increasing end user satisfaction. Access-eGov [14] is an EC research project conducting research in the personalization area.

- Interoperability – as discussed and proposed by the European Commission in the European Interoperability Framework (EIF) [9] and outlined in the analysis by Luis Guijarro [10]. Some of the EC research projects targeting this issue include eMayor [11], R4eGov [12] and Terregov [13] project.
- Quality Assurance – provision of the Quality of Service (QoS) aspects to the E-Government services plays major role for selection of the most appropriate service delivery channels and service providers. QUALEG [15] is an EC research project tackling this issue. Further interesting aspects are research conducted by Magoutas, et al. on QeGS [16] and e-GovQoS [17] ontology based approaches for QoS in E-Government service delivery.
- Knowledge Management – involvement of KM in provision of E-Government services as described by Lopez, et al. [18] and Chung, et al. [19] in order to support the knowledge intensive tasks is becoming more important due to high complexity of such processes and amount of hidden knowledge both for the end user as well as for service providers. Work done by Mentzas, et al. [20] and Hinkelmann, et al. [21] provides a starting point for the semantic architecture used to allow provisioning of the knowledge services in the E-Government area.
- Adaptivity – design and development of adaptive service delivery processes, capable of adapting themselves to the changes in the environment without changing the process itself. Research in this area was carried out in the already mentioned eMayor and FIT EC Research Project [22].

Although there are much more different research topics in this area, the aforementioned topics have been selected due to their relevance for the work presented in the following two chapters.

III. E-GOVERNMENT SEMANTIC WORKFLOWS

Semantic workflows in the scope of the research projects presented within this paper are defined as workflows enriched with semantic technologies in order to provide answers to E-Government scenario specific challenges

The need for deployment of the such workflows in the area of public administration rose from different requirements. Mapped on the abovementioned research topics, the main goals were to offer: (1) Personalization to make the delivery of the service much more user friendly and cost effective [39], (2) Interoperability without the need to exchange the existing systems, but to make them interoperable through usage of the semantic technologies, (3) Quality of Service on one side to support the Personalization initiative and on the other hand to allow assessment of the available E-Government services, (4) Knowledge Management to support knowledge provision for knowledge intensive tasks and (5) Adaptivity to allow rapid responses on the changes in the environment without the need to apply dramatically changes to the deployed processes. In the

following, two application scenarios within two EC research projects for semantically enriched E-Government workflows (Chapter III A) and their implementation (Chapter III B) are presented.

A. Application Scenarios

The first application scenario for semantic E-Government workflows carried out in the FIT EC research project (IST-2004-27090) is concerned with the development and deployment of the so-called “adaptive” workflows. FIT aims to “develop, test and validate a self-adaptive e-Government framework based on semantic technologies that will ensure that the quality of public services is proactively and continually fit to the changing preferences and increasing expectations of e-citizens” [22]. Research conducted within this project tackled the issue that it is not easy to apply fast changes to the deployed E-Government processes although they are used in a changing environment. The answer provided was to make the implicit Business Rules hidden in the processes explicit, therefore transferring the need to comply with changes to the rule level and eliminating the need to constantly change and redeploy the business processes/ workflows. The second application scenario presented in this paper tackles the issue of providing “smart” workflows. That is, based on the personalization and quality of services requirements and considering the interoperability layer, workflows are made executable at the runtime according the end users preferences. This research challenge has been conducted within the LD-CAST EC research project (IST-2004-26919). The project aims to “specify, develop and evaluate an integrated network of seamless interoperable public services to businesses in the context of Enlarged Europe” [23].

B. Implementation of Adaptive and Smart Workflows

The starting point for both projects (LD-CAST and FIT) was the modelling of the relevant E-Government processes conducted by teams consisting of experienced modellers and domain experts. In case of LD-CAST the domain knowledge was provided by representatives of European Chambers of Commerce and in case of FIT by representatives of an Austrian municipality and a Greek ministry taking part in the project. Modelling was carried out using the ADONIS[®] [24] business process management tool and the especially developed E-Government modelling language ADOeGov[®] [25] which extended the standard modelling language provided with ADONIS[®] by providing E-Government specifics (e.g. top-down based life-event approach). ADOeGov[®] was extended during the projects [34] to include BPEL [26] notation to enable orchestration of workflow, OWL [27] notation in order to be able to establish loose coupling between E-Government processes, abstract- and executable workflows through automatic ontology generation and SWRL [38] notation to allow the transformation of semi-formal business rules to formal and thus executable rules.

In order to implement smart workflows in LD-CAST a four-phase methodology was followed: (1) Modelling of the E-Government processes and generation of the

corresponding abstract workflows. Workflows are marked as abstract as they don't have any services bound to the activities until they are transferred to the runtime environment; (2) automatic generation of the LD-CAST ontology out of the modelled E-Government processes, using the ADOeGov[®] OWL export functionality. Ontology was then enriched using the OPAL [28] framework for building business based ontologies; (3) registration of the concrete services by service providers used for creating executable workflows at the runtime; and (4) semantic linking of the E-Government processes, abstract workflows and concrete services through annotation with the corresponding concept available in the LD-CAST ontology.

At runtime an end user selects an appropriate E-Government process to be executed and based on the selected QoS parameters the LD-CAST portal provides possible executable workflows.

To configure adaptive workflows for the back-office of the system in FIT the following methodology was followed: (1) Semi-formal business processes and rules were modelled by the domain experts using as input their domain knowledge, document sources (e.g. laws, regulations etc.), and existing process models. (2) Business rules defined are grouped to rule-sets and assigned to the corresponding activities, decisions or processes. (3) Using the semi-formal business processes and rules as input, a transformation into an executable representation takes place, where semi-formal business processes are transformed into BPEL format and business rules are transformed into OWL and SWRL.

At runtime an adaptive workflow is triggered, e.g. when a citizen submits an application for building permission. In the back-office the public administration employees process this application. The tasks to be executed are provided by the workflow engine, which calls the business rules engine to determine activities and processes to be executed during process runtime (Variable Process Execution), to select employees based on special skills, to present information depending on user categories or to select a particular Web-Service (Intelligent Resource Allocation), to control the process flow (Intelligent Branching and Decision Making) or to evaluate mandatory constraints or guidelines (Constraint Checking).

After integration of the results with the end user systems technical, organizational and social aspects were evaluated. The approach was judged as to increase the efficiency, effectiveness and flexibility of the system. Details on the evaluation are provided in the related report [37].

Although the introduction of semantics enables adaptive and smart workflows, on demand knowledge provision can further improve the provision of public services.

IV. SERVICE ORIENTED KNOWLEDGE PROVISION

The main idea behind service oriented knowledge provision (SOKP) is the goal to provide compliance between the traditional knowledge provision and the fact that many of the ICT systems are either deployed already using the SOA approach or are being directed towards the SOA approach. The goal followed in the E-Government area is to pipe the available knowledge towards the knowledge intensive tasks

in order to support equally the public administrations as service providers in carrying out the tasks, as well as the end users accessing the service.

The SOKP approach assumes that (1) knowledge is hybrid – implicit and explicit, (2) knowledge is implemented in tools for service delivery and (3) knowledge needs to be virtualized. SOKP is built around the principle that an end user supported by the knowledge does not need to care about where this knowledge comes from or about the form of the knowledge source, but just to send query's to the system (either push-based when executing a specific task, or pull-based requesting information before execution) and receive the requested information.

Another important aspect is that the knowledge provision is carried out at an agreed time and in agreed quality (QoS principle) and that the knowledge is presented in the domain language of the end user (Interoperability principle).

In the following the conceptual architecture of a SOKP system based on the Knowledge Bus is presented (Chapter IV A) and a prototype application of a SOKP system is presented (Chapter IV B).

A. *The Knowledge Bus as a Basis for Service Oriented Knowledge Provision*

This section introduces the concept of the knowledge bus, which is depicted in Figure 1. The ultimate goal of the knowledge bus is to support the knowledge request of a knowledge worker for a certain application scenario. In order to bridge the gap between knowledge demand at the application scenario layer and knowledge provision at the knowledge working and usage layer the PROMOTE[®] [33] approach is used to design all necessary aspects of knowledge management. The current enterprise service bus [29] technology is used as technical basis, as it provides functionalities like e.g. mediations, messaging, communication, and security.

The enterprise service bus will be configured with PROMOTE[®] using process models that are stored in the knowledge flow registry. The traditional service registry is enriched with an ontology based on the knowledge management service classification [34] that enables a semantic service registry. Additional to the usual enterprise service bus configuration, PROMOTE[®] provides a skill-based authorization repository that describes knowledge workers, their competences, roles and topics of interest.

One of the challenging tasks is the virtualization of knowledge. Knowledge services are used to access the virtualized knowledge sources. These services may be based on explicit knowledge sources such as databases, business models, etc. and implicit knowledge sources such as human domain experts (e.g. Skype Prime service [30]).

The core functionality of the Knowledge Bus is to provide the messaging system for orchestration and retrieval of knowledge services. It is thereby taking over the responsibility to deliver the appropriate knowledge to the requestor, by allowing queries. Due to different complexity levels of the query the Knowledge Bus is either providing answers with one to one mapping with the available knowledge services, or by orchestrating different knowledge

services in order to retrieve the knowledge required by the requestor.

Another important aspect in orchestrating/provision of the knowledge services is to agree on the time and quality, thus the (Knowledge) Service Level Agreement.

Figure 1 represents a possible deployment scenario for the Knowledge Bus divided into three conceptual layers: (1) Knowledge Work and Usage Layer – containing the ESB based technical basis, (2) Knowledge Management Layer – containing the Knowledge Bus and (3) Application Scenario Layer – application scenario for the specific use-case.

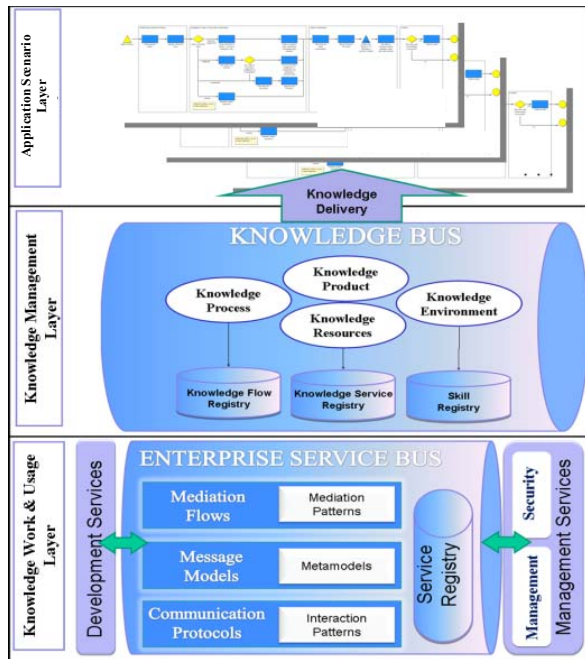


Figure 1. The Knowledge Bus concept

The Knowledge Bus is the technical core element of the E-Learning EC project MATURE [33] and will be implemented as a proof of the scientific concepts to show the added value in practice.

B. Service Oriented Knowledge Provision in BREIN

The EC research project BREIN [30] provided an initial proof of concept of the Knowledge Bus concept. In this Grid based research project the concept was applied to ensure knowledge transfer and sustainability. The goal was to provide external developers with model-driven guidance through the development of new or the extension of existing applications and services for the BREIN platform.

Figure 2 presents five layers, where each of these layers is seen as an entry point to access knowledge. The bottom layer - the resource layer - collects all knowledge resources (e.g. videos, code samples, documents). Every additional layer above provides smarter access to knowledge.

When the user accesses the knowledge on the top layer – the roadmap layer – a profiling service will define the role of the user. Based on this classification the user is supported by the appropriate knowledge services, which provide him/her

with the required knowledge to fulfil the desired goal (e.g. to develop a new application for BREIN).

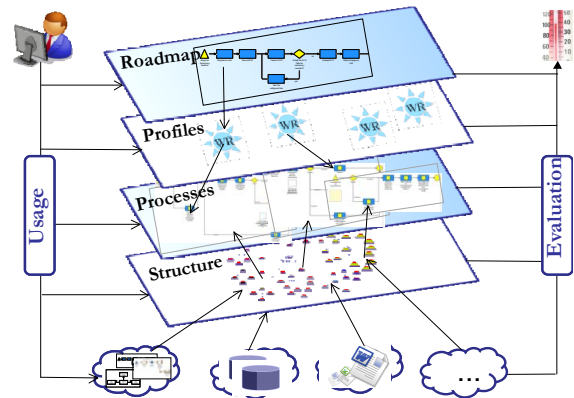


Figure 2. Service oriented knowledge provision in BREIN

As the figure 2 depicts, the service provision is evaluated by gathering user feedback in order to identify potentials for further improvement of the QoS of the provided knowledge services. Details on the service oriented knowledge provision in BREIN can be found in [31] and [32].

V. CONCLUSION AND OUTLOOK

This paper provided an overview of research in the area of semantic E-Government workflows and provided examples how to enhance this workflows by introducing a service oriented approach for knowledge provision (SOKP). Initial prototypes introduced first steps in developing a holistic framework for this approach.

The introduced Knowledge Bus will be implemented in the EC project MATURE, where it is seen as an interface for accessing various knowledge sources and for registry, discovery and invoking of services as well as messaging between them and will be applied in personal and organizational learning environment.

Further research on the identified challenge of interoperability of the domain knowledge expressed in different notations/native languages will be done in the upcoming EC research Project plugIT starting in 2009.

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